

- 4 The photograph shows a young giant panda feeding on bamboo.



(Source: © Cheryl Schneider/Alamy Stock Photo)

Giant pandas have evolved the ability to eat bamboo.

Genes that code for enzymes that digest the cellulose in the bamboo are not present in the genome of the giant panda.

The giant pandas depend on microorganisms in their intestines to produce enzymes that can digest the cellulose.

- (a) Which type of glycosidic bonds are broken in the digestion of cellulose?

(1)

- ☐ **A** α 1-4
- ☐ **B** α 1-6
- ☐ **C** β 1-4
- ☐ **D** β 1-6

- (b) The microorganisms in the intestines of giant pandas include both bacteria and fungi.

Fungi are eukaryotic organisms.

Put one cross [x] in the appropriate box, in each row, to show if genes are found in each of the structures in these microorganisms.

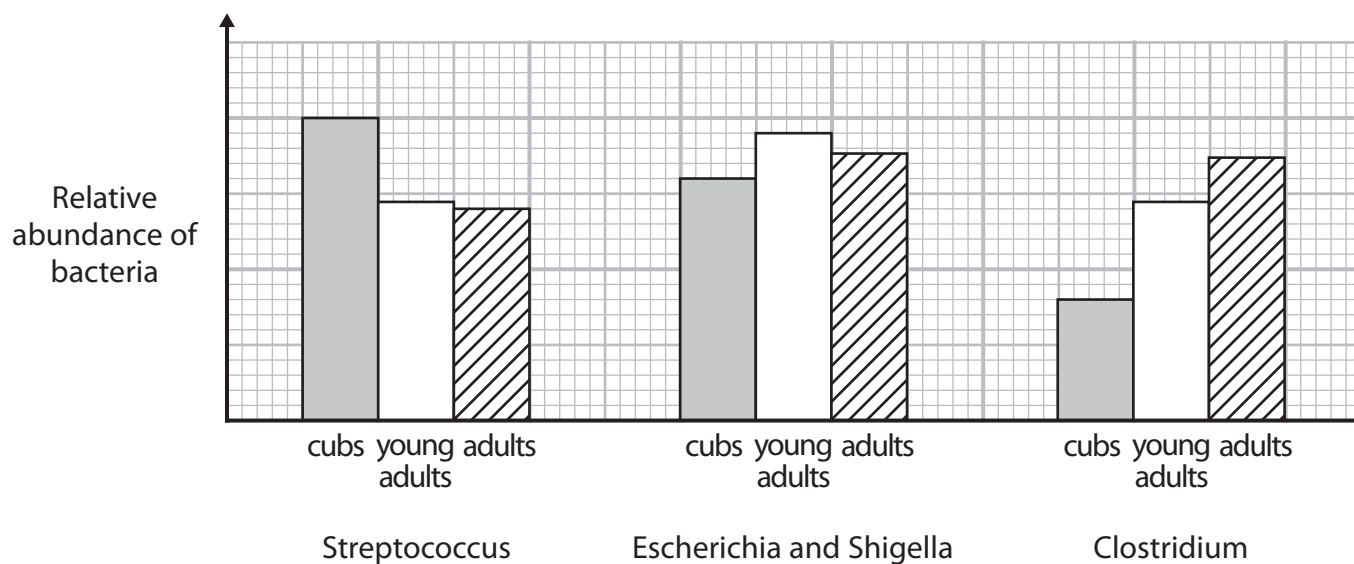
(3)

Structure	Microorganism			
	Bacteria only	Fungi only	Both bacteria and fungi	Neither bacteria nor fungi
mitochondria	☐	☐	☐	☐
nuclei	☐	☐	☐	☐
70S (small) ribosomes	☐	☐	☐	☐

(c) In one study, the succession of bacteria in the intestines of developing giant pandas was determined.

Three stages of development were studied: cubs, young adults and adults.

The graph shows the relative abundance of four types of bacteria in the three developmental stages of the giant panda.



(i) Which sequence shows the succession of bacteria during the development of a giant panda?

(1)

- ☐ **A** Clostridium → Streptococcus → Escherichia and Shigella
- ☐ **B** Escherichia and Shigella → Clostridium → Streptococcus
- ☐ **C** Escherichia and Shigella → Streptococcus → Clostridium
- ☐ **D** Streptococcus → Escherichia and Shigella → Clostridium

- (ii) Describe one method that could be used in this study to determine the relative abundance of *Streptococcus* in a giant panda.

(4)

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(Total for Question 4 = 9 marks)
